A PROJECT REPORT

ON

APPLICATIONS OF MINIATURE STIRLING CRYOCOOLER



Submitted by

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CERTIFICATE

This is to certify that this report entitled "Applications of Miniature Stirling Cycle Cryocooler" embodies the work done by Shreya Narayanan in partial fulfilment of B.Tech. (Mechanical Engineering with Electric Vehicles), in the "Cryogenic Lab" of "Engineering Services Division", Solid State Physics Laboratory, Defence Research and Development Organization (DRDO), Delhi from 26th June to 4th August 2023 The project is an original work of the candidate and was completed successfully in time.

The candidate had a good code of conduct and sincerity towards her tasks during the course of the training.

I wish her success in all her future endeavours.

ACKNOWLEDGEMENT

While presenting this report I would like to express my deep sense of gratitude towards **Dr. Seema Vinayak, Sc 'H', Director, Solid State Physics Laboratory**, for allowing me to carry out my training in this establishment and making it a truly memorable one.

I am also grateful to my mentor **Sh. Gaurav Pratap, Sc 'E'** who was an indispensable part of my training giving me unending guidance, inspiration, encouragement and providing me a conductive environment throughout my training at D.R.D.O, Delhi. The training was an extremely productive and enriching experience, not only technically but also in the matter of providing some practical skills.

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ABSTRACT

A miniature Stirling cryocooler is a type of cryogenic refrigeration system based on the Stirling cycle, designed to achieve low temperatures. Stirling cryocoolers operate using the principles of cyclic compression and expansion of gas to transfer heat from one location to another, ultimately achieving cooling effects. Miniature Stirling cryocoolers have the advantage of being more compact and lighter compared to other cryocooler types, making them suitable for portable and space-constrained applications. However, they also have certain limitations, such as higher manufacturing costs and lower cooling capacities compared to larger cryocoolers like Gifford-McMahon or Pulse Tube cryocoolers.

ABOUT THE DRDO

Defence Research and Development Organization (DRDO) was established in 1958 by amalgamating Defence Science Organization and some of the technical development establishments. A separate department of Defence Research and Development was formed in 1980, which now administers DRDO and its 51 laboratories and establishments.

The department of Defence Research and Development formulates and executes programmes of scientific design and development in fields of relevance to national security leading to the induction of new weapons, platforms and other equipment required by the armed forces. It also functions as the nodal agency for execution of the major development programmes of relevance to Defence through integration of research, development testing and production facilities with the national scientific institutions, public sector undertakings and other agencies. Research and Development activities at DRDO cover important demarcated disciplines like aeronautics, rockets and missiles, electronics and instrumentation, combat vehicles, engineering , naval systems, armament technology including explosive research, terrain research, advanced computing, artificial intelligence, robotic systems technology and nuclear medicine.

Several high technology projects are in various stages of development. The main battle tank Arjun incorporating state-of-art technologies with superior firepower, high mobility and excellent protection has been developed. A limited number of tanks are being produced as pre-production series which are in final stages of evaluation by the army.

The integrated guided missile development program comprises of five missile systems, Prithvi, surface to surface tactical battlefield, Akash, medium range surface-to-air missile system. Trishul, A short range surface-to-air missile and Agni, an intermediate range ballistic missile.

ABOUT THE SSPL

Solid State Physics Laboratory (SSPL), one of the establishments under the Defence R&D Organization (DRDO), Ministry of Defence, was established in 1962 with the broad objective of developing an R&D base in the field of Solid State Materials, Devices and Sub-systems. The Laboratory has a vision to be the centre of excellence in the development of Solid State Materials, Devices and has a Mission to develop and characterize high purity materials and solid state devices and to enhance infrastructure, technology for meeting the futuristic challenges.

From a small beginning, the laboratory has grown into premier Research and Development establishment in the field of electronic devices and is well known both in the country and abroad. The nature of activities of SSPL are interdisciplinary involving Physics, Chemistry, Material Science and Mechanical Engineering Science. The laboratory draws its strength from a pool of well qualified scientists and engineers from various disciplines. The laboratory employs about 600 personnel of which more than 375 are scientists/engineers. Since its inception, the laboratory has been continuously updating its capabilities, so as to keep pace with phenomenal development in the field of semiconductor materials and devices based on Silicon. Over the years the work has reached its present day level based on conductors, piezoelectric materials and ferrite and ferroelectric materials.

CHAPTER1: INTRODUCTION

1.1 CRYOGENICS



Fig 1.1: Liquid Nitrogen
Ref: www.thoughtco.com

Cryogenics is defined as the scientific study of materials and their behaviour at extremely low temperatures. The word comes from the Greek *cryo*, which means "cold", and *genic*, which means "producing". The term is usually encountered in the context of physics, materials science, and medicine. A scientists who studies cryogenics is called a *cryogenicist*. A cryogenic material may be termed a *cryogen*. Although cold temperatures may be reported using any temperature scale, the Kelvin and Rankine scales are most common because they are absolute scales that have positive numbers.

1.2 CRYOGENIC FLUIDS

Cryogenic liquids are typically stored in devices called Dewar flasks. These are double-walled containers that have a vacuum between the walls for insulation. Dewar flasks intended for use with extremely cold liquids (e.g., liquid helium) have an additional insulating container filled with liquid nitrogen. They are named for their inventor, James Dewar. The flasks allow gas to escape the container to prevent pressure build up from boiling that could lead to an explosion.

The following fluids are most often used in cryogenics:

Fluid	Boiling Point (K)
Helium-3	3.19
Helium-4	4.214
Hydrogen	20.27
Neon	27.09
Nitrogen	77.36
Air	78.8
Fluorine	85.24
Argon	87.24
Oxygen	90.18
Methane	111.7

1.3 CRYOCOOLERS

1. A refrigerator designed to reach cryogenic temperatures is often called a cryocooler. The term is most often used for smaller systems, typically table-top size, with input powers less than about 20 kW. Some can have input powers as low as 2-3 W.
2. Large systems, such as those used for cooling the superconducting magnets in particle accelerators are more often called cryogenic refrigerators. Their input powers can be as high as 1 MW.
3. In most cases cryocoolers use a cryogenic fluid as the working substance and employ moving parts to cycle the fluid around a thermodynamic cycle. The fluid is typically compressed at room temperature, precooled in a heat exchanger, then expanded at some low temperature. The returning low-pressure fluid passes through the heat exchanger to precool the high-pressure fluid before entering the compressor intake. The cycle is then repeated.

There are six commonly used cryocooler types, which are divided into two categories — recuperative cycles and regenerative cycles.

They are then divided into three cycles of recuperative and regenerative each:

- Joule – Thomson
- Braton
- Claude
- Stirling
- Pulse – Tube
- Gifford – McMahon

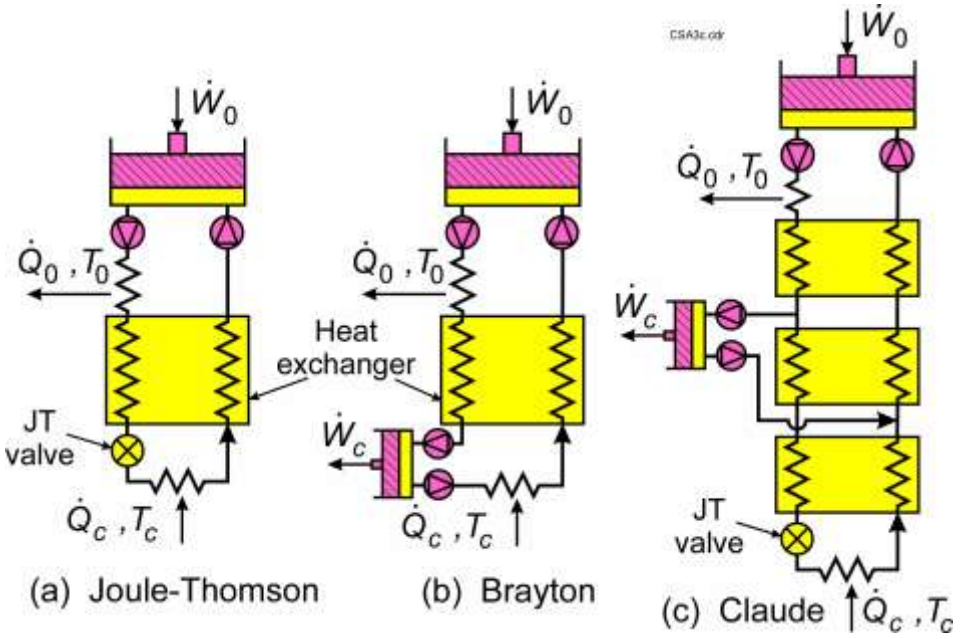


Fig 1.2 a): Recuperative cycles
Ref: https://trc.nist.gov

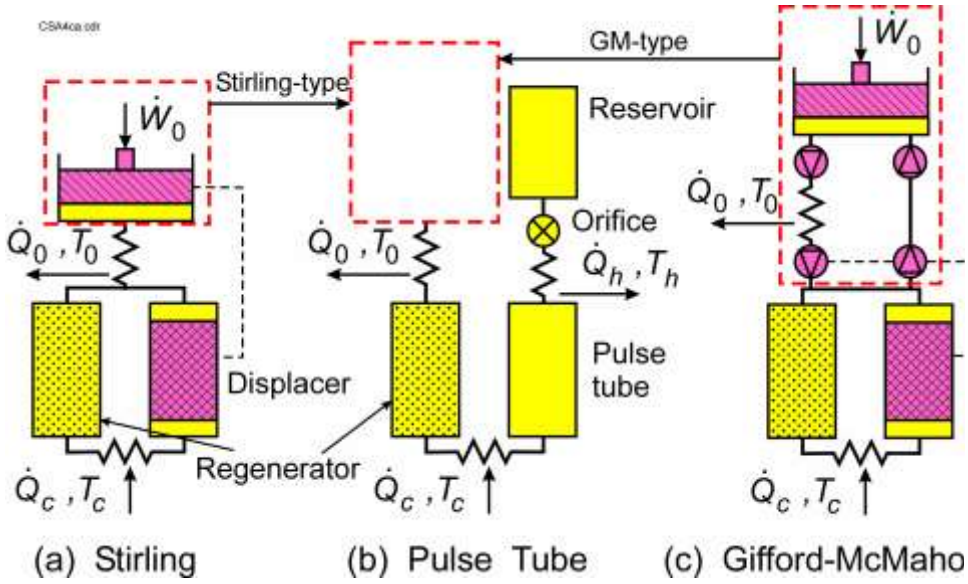


Fig 1.2 b): Regenerative cycles
Ref: https://trc.nist.gov

CHAPTER 2: MINIATURE STIRLING CRYOCOOLERS

Stirling Cryogenerators are based on the reversed-Stirling cycle. These Cryogenerators, also known as Cryocoolers or Cryo-refrigerators, are named SPC's (Stirling Process Coolers) and are the heart and key component in all that Stirling does.

Stirling Cryocoolers have a number of unique features:

1. Highest efficiency possible (through reversed Stirling Cycle technology) leading to low electricity consumption
2. Fully automated and PLC controlled
3. Process specific heat exchanger integrated in the cold head
4. Flexible process integration
5. Robust design with proven reliability: some Cryocoolers are over 40 years in operation
6. All rotational and reciprocating seals work at ambient temperatures
7. Maintenance friendly
8. Annual inspection and cleaning can be done by customer's engineer at site
9. Worldwide service available; mostly with nearby local service engineer
10. Various options available such as capacity control, ATEX, maritime version etc.
11. Supply according to all applicable coding, power grid and requirements of the customers' site.

2.1 STERLING CYCLE

1. Stirling coolers are based on causing a working gas to undergo a Stirling cycle which consists of two constant volume processes and two isothermal processes.
2. Devices consist of a compressor pump and a displacer unit with a regenerative heat exchanger, known as a regenerator.
3. Stirling engines require several minutes of cool-down time; the working fluid is helium. The development of two-stage devices extend the lower temperature range from 60–80 to 15–30 K.
4. Both Joule–Thompson and engine-cooled detectors are housed in precision-bore dewars into which the cooling device is inserted (see Fig.). The detector, mounted in the vacuum space at the end of the inner wall of the dewar and surrounded by a cooled radiation shield compatible with the convergence angle of the optical system, looks out through an IR window.
5. In some dewars, the electrical leads to detector elements are embedded in the inner wall of the dewar to protect them from damage due to vibration. After 3000 to 5000 h of operation, Stirling cooler requires factory service to maintain performance.

6. Because the dewar and cooler is one IDA unit, the entire unit must be serviced together.

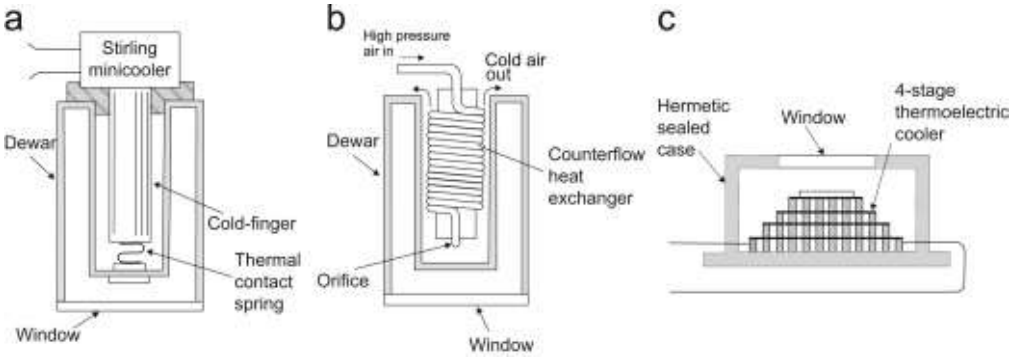


Fig 2.1: Three ways of cooling IR detectors: **a)** Stirling-cycle engine, **b)** Joule–Thompson cooler, and **c)** four-stage thermoelectric cooler (Peltier cooling).
Ref: www.sciencedirect.com

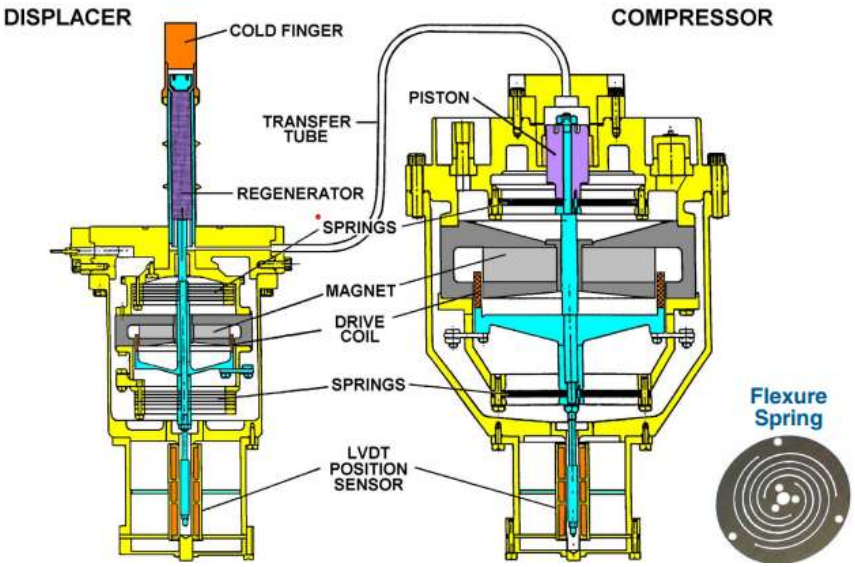


Fig 2.2: Stirling cycle
Ref: www.jlab.org

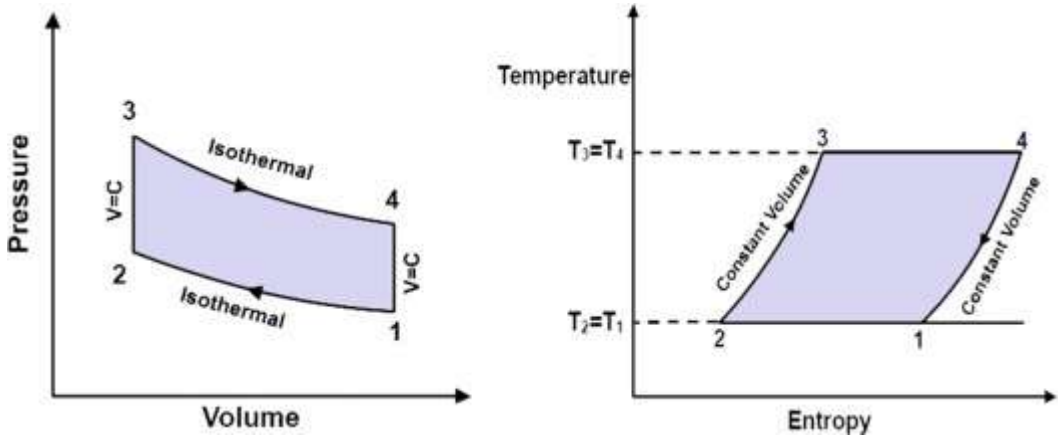


Fig 2.3: Stirling cycle plot
Ref: www.sciencedirect.com

2.2 SIZE EFFECTS ON THE PERFORMANCE

Size effects on the performance of Stirling cycle cryocoolers were investigated by examining each individual loss associated with the regenerator and combining these effects. For the fixed cycle parameters and given regenerator length scale, it was found that only for a specific range of the hydrodynamic diameter the system can produce net refrigeration and there is an optimum hydraulic diameter at which the maximum net refrigeration is achieved. When the hydraulic diameter is less than the optimum value, the regenerator performance is controlled by the pressure drop loss; when the hydraulic diameter is greater than the optimum value, the system performance is controlled by the thermal losses.

It was also found that there exists an optimum ratio between the hydraulic diameter and the length of the regenerator that offers the maximum net refrigeration. As the regenerator length is decreased, the optimum hydraulic diameter-to-length ratio increases; and the system performance is increased that is controlled by the pressure drop loss and heat conduction loss. Choosing appropriate regenerator characteristic sizes in small-scale systems are more critical than in large-scale ones.

CHAPTER 3: IMPLEMENTATION

There are a number of implementations but a few are discussed below.

3.1 INFRARED SENSORS AND DETECTORS

1. Four strokes are combined in the Stirling cycle: compression, expansion, and two isochoric transformations.
2. Stirling coolers are mechanical machines built to apply the Stirling cycle to a working fluid. To do so, they combine two movements:
   - reciprocating a motion of a compressor piston alternating compression and expansion phases of the gas present in the working chamber. This stands for two strokes of the reversed Stirling cycle.
   - reciprocating a movement of a displacer allowing transfer of a gas volume from cold to hot side to perform two remaining isochoric strokes of the Stirling cycle. As the gas travels from the cold side to the hot side and back, the gas passes through a regenerator, a heat exchanger, to emphasize the cooling effect.

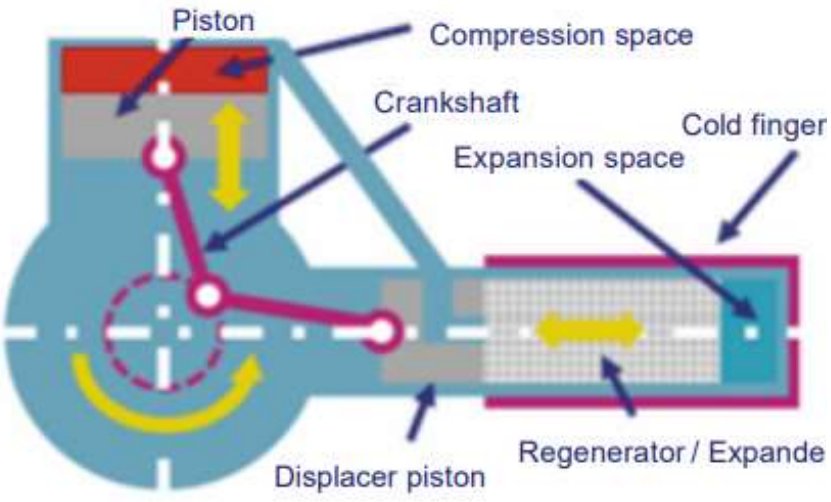


Fig 3.1: Rotary Stirling cycle
Ref: www.ricor.com

Stirling coolers are closed-circuit regenerative engines and use helium gas as the working fluid. There is no contact between helium and parts to be cooled and there is no phase transformation in this process. Such system can be miniaturized to fit size weight and power requirements of most of IR detection applications. There are two main types of Stirling cryocoolers: linear and rotary. Working principles are globally the same, the difference lies in the actuation mode of the compressor piston:

• in a linear cooler, the compressor piston is placed in a coil and actuated in a reciprocating movement by ways of electro-magnetism.

• in a rotary cooler, there is a motor with a rotating shaft, the compressor piston is actuated from the shaft using a conrod – crankshaft assembly.

3.2 SPACE APPLICATIONS

Space applications of miniature Stirling cryocoolers are essential for cooling various scientific instruments and detectors used in space missions. These cryocoolers play a crucial role in enabling high-resolution imaging, spectroscopy, and other scientific measurements in the extreme conditions of outer space. Here are some specific space applications of miniature Stirling cryocoolers:

1. Infrared Astronomy: Space telescopes used for infrared astronomy require very low temperatures to reduce the background thermal radiation and improve the sensitivity of infrared detectors. Miniature Stirling cryocoolers are used to cool these detectors, allowing scientists to study distant galaxies, stars, and other celestial objects in the infrared spectrum.
2. X-ray Astronomy: Some X-ray detectors in space-based telescopes and observatories operate most efficiently at cryogenic temperatures. Stirling cryocoolers are employed to cool these detectors and enhance their sensitivity, enabling the study of X-ray emissions from high-energy cosmic sources.
3. Spectroscopy: Spectrometers used in space missions for analyzing the composition of planetary atmospheres or interstellar gases often require cryogenic cooling. Miniature Stirling cryocoolers provide the necessary cooling to achieve the required temperature stability for precise spectroscopic measurements.
4. Cosmic Microwave Background (CMB) Studies: The CMB is the afterglow of the Big Bang and provides valuable information about the early universe. Instruments studying the CMB, such as space-based telescopes and interferometers, use miniature Stirling cryocoolers to cool their detectors to extremely low temperatures.
5. Space-Based Interferometry: Interferometers used for precise measurements of astronomical objects, such as space-based optical interferometers or gravitational wave detectors, often rely on cryogenic cooling to stabilize their optical paths and improve their accuracy. Miniature Stirling cryocoolers can provide the required cooling for such systems.
6. Space-Based Lidar Systems: Some lidar systems in space missions utilize cryogenic cooling to cool their detectors, allowing precise measurements of atmospheric parameters, such as aerosols and greenhouse gases.
7. Space Spectrographs: Spectrographs in space telescopes and observatories are used to study the spectra of distant objects, including stars, exoplanets, and galaxies. Cryogenic cooling from Stirling cryocoolers ensures the stability and performance of these instruments.
8. Space-Based Gravitational Wave Detectors: Future space-based gravitational wave observatories may utilize cryogenic cooling to reduce thermal noise and improve the sensitivity of their detectors for detecting gravitational waves.
9. Space Exploration Instruments: Various scientific instruments on spacecraft and planetary rovers may require cooling for their detectors and sensors to conduct detailed analyses of planetary surfaces, atmospheres, and environments.

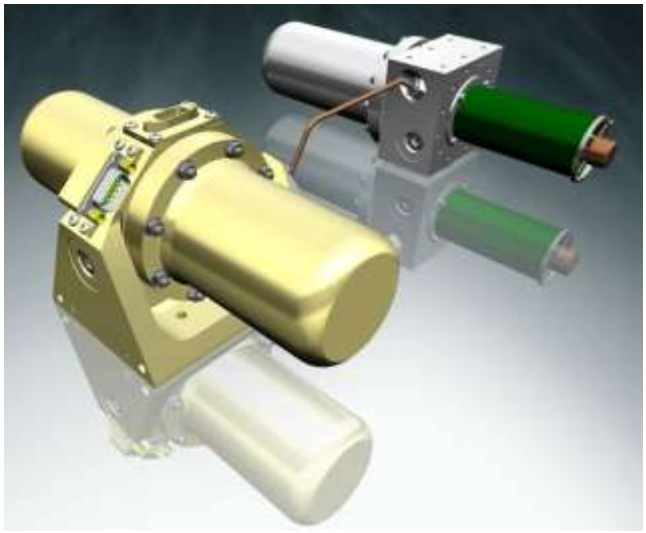


Fig 3.2: Cryogenics used in space
Ref: www.openaccessgovernment.org

3.3 ELECTRONICS COOLING

Electronics cooling is a critical aspect of electronic device design and operation. As electronic components become more powerful and compact, they generate more heat, which can negatively impact their performance and reliability. Effective cooling is necessary to dissipate this heat and maintain the components at safe operating temperatures. Miniature Stirling cryocoolers offer a unique and efficient solution for electronics cooling, especially for high-temperature superconducting devices and other applications that require cryogenic temperatures. Here's how Stirling cryocoolers are used for electronics cooling:

1. High-Temperature Superconducting (HTS) Devices
2. Quantum Computing
3. High-Performance Computing (HPC)
4. Radar Systems
5. Power Electronics
6. Laser Diodes and Photodetectors
7. Radio Astronomy Receivers
8. High-Speed Electronics

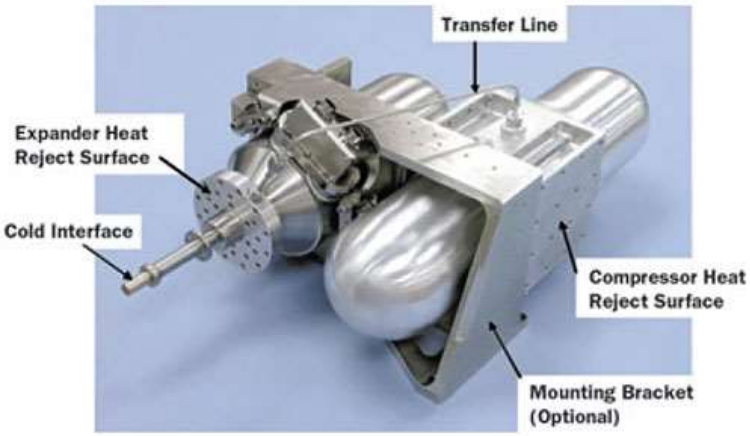


Fig 3.3: Cryocooler in Laptop
Ref: www.jlab.org

3.4 CRYOPRESERVATION

In medical and biotechnology applications, Stirling cryocoolers are used for cryopreservation, where biological samples, tissues, and cells are stored at very low temperatures to maintain their viability for extended periods.



Fig 3.4: Embryo Cryopreservation
Ref: www.hopkinsmedicine.org



Fig 3.5: Time and Death pause for living humans
Ref: www.dailysabah.com

3.5 RESEARCH AND SCIENTIFIC INSTRUMENTS

Stirling cryocoolers are used in various scientific research instruments like spectrometers, mass spectrometers, and cryogenic sensors where stable and low-temperature conditions are required.



Fig 3.6: Cryocooler used in Laboratory research
Ref: https://link.springer.com

3.6 CRYOGENIC COOLING FOR COOLING CHAMBERS

Miniature Stirling cryocoolers are used in cooling chambers to create low-temperature environments for testing and calibrating electronic components, sensors, and materials.

3.7 COOLING FOR SUPERCONDUCTING DEVICES

Cooling for superconducting devices is a critical aspect of their operation because these devices exhibit unique properties at very low temperatures. Superconductors are materials that, below a certain critical temperature, have zero electrical resistance and can expel magnetic fields—a phenomenon known as the Meissner effect. To achieve and maintain these superconducting states, superconducting devices require cryogenic cooling. Miniature Stirling cryocoolers are employed to provide the necessary cooling for various superconducting applications. Here's how cooling with Stirling cryocoolers benefits superconducting devices:

1. Zero Resistance Operation: Superconducting devices, such as superconducting qubits in quantum computing or superconducting magnets in magnetic resonance imaging (MRI), require cryogenic cooling to operate in their superconducting states. When cooled below their critical temperature, these devices can conduct electricity without any resistance, leading to increased efficiency and sensitivity.
2. Quantum Computing: Quantum computing relies on superconducting qubits that need to be cooled to extremely low temperatures (millikelvin range) to maintain their quantum coherence and stability. Stirling cryocoolers can cool the quantum processors efficiently, allowing for reliable quantum operations and minimizing decoherence caused by thermal fluctuations.
3. Magnetic Resonance Imaging (MRI): MRI machines utilize superconducting magnets that generate strong and stable magnetic fields for imaging. Cryogenic cooling is essential to maintain the superconducting state of these magnets, enabling consistent and high-resolution imaging.
4. Particle Accelerators: Superconducting magnets are often used in particle accelerators, such as those in high-energy physics experiments or medical particle therapy systems. Cooling these magnets to cryogenic temperatures reduces electrical losses and enhances the magnetic field strength, contributing to improved accelerator performance.
5. High-Field Science: In various scientific research fields like materials science and condensed matter physics, superconducting magnets are used to generate extremely high magnetic fields. Cooling these magnets with Stirling cryocoolers allows researchers to reach the necessary low temperatures for their experiments.
6. Electric Power Transmission: Superconducting materials have the potential to revolutionize power transmission due to their ability to carry large currents with minimal loss. Cooling these superconducting power cables or fault current limiters is essential to maintain superconducting properties during operation.
7. Particle Detectors: Superconducting detectors, such as transition-edge sensors (TES) or superconducting nanowire single-photon detectors (SNSPDs), can be used for

sensitive measurements in fields like astronomy and particle physics. Stirling cryocoolers provide the cooling required to operate these detectors efficiently.

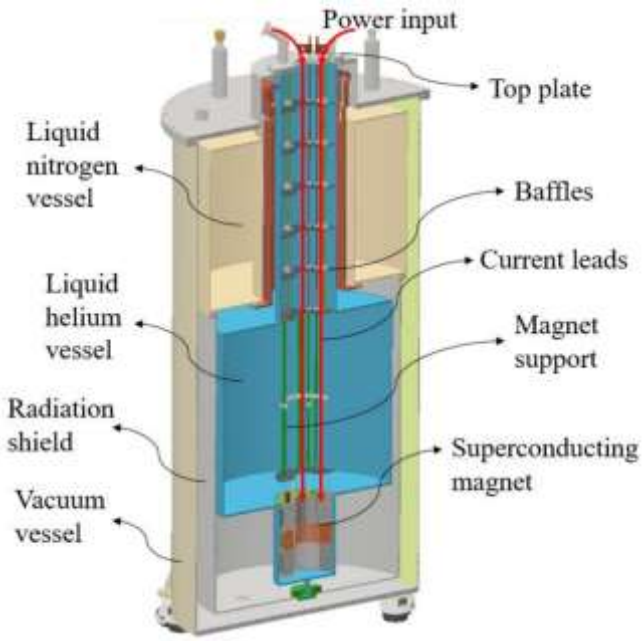


Fig 3.7: Cryocooler for superconduction
Ref: www.mdpi.com

3.8 CRYOGENIC COOLING FOR OPTICAL DEVICES

In some high-precision optical systems, such as laser stabilization or quantum optics experiments, low temperatures are essential to achieve specific performance characteristics. Miniature Stirling cryocoolers can provide the necessary cooling in such applications.

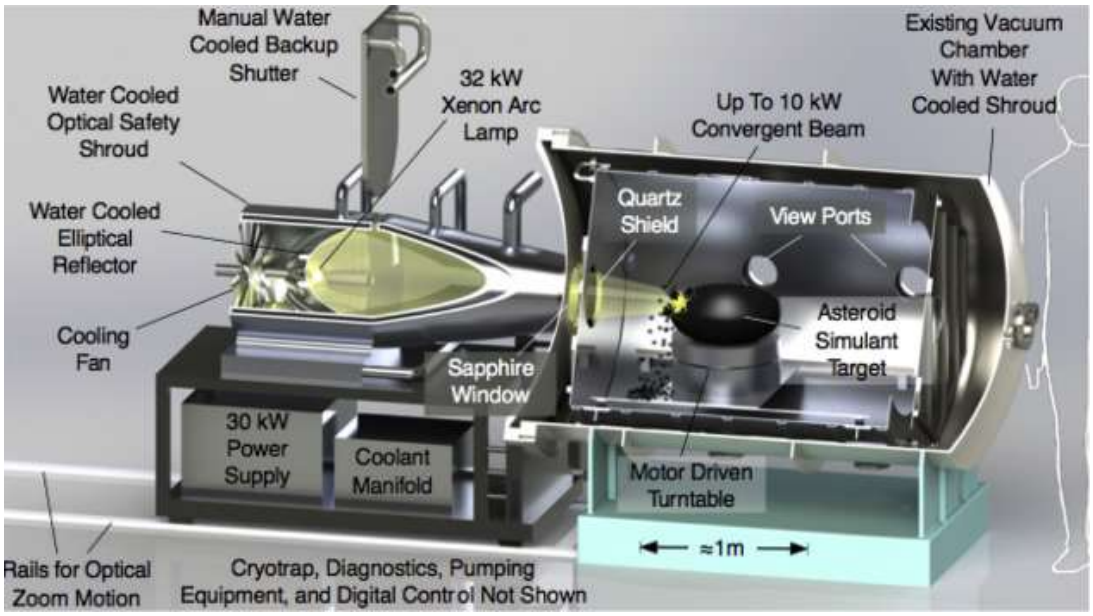


Fig 3.8: Cryocooler for optical devices
Ref: www.sciencedirect.com

3.9 MILITARY APPLICATIONS

Miniature Stirling cryocoolers have several important military applications, especially in the defence and security sectors. Their compact size, low vibration, and reliable cooling capabilities make them well-suited for various military applications. Here are some key military applications of miniature Stirling cryocoolers:

1. Infrared Countermeasures: Stirling cryocoolers are used to cool infrared (IR) sensors and detectors in missile warning systems and aircraft countermeasures. By cooling the IR sensors, these systems can detect and track heat-seeking missiles and threats more effectively.
2. Target Acquisition Systems: Miniature Stirling cryocoolers are employed in target acquisition systems for military reconnaissance and surveillance. These systems use cooled IR detectors to identify and track targets, including vehicles, personnel, and other objects of interest.
3. Night Vision Devices: Stirling cryocoolers are utilized in night vision goggles, monoculars, and binoculars. Cooling the IR sensors in these devices enhances their sensitivity, enabling military personnel to see in low-light conditions and at night.
4. Infrared Guided Missiles: Stirling cryocoolers cool the IR seekers in infrared-guided missiles, allowing them to home in on the heat emitted by their targets. This makes these missiles highly effective against armored vehicles and aircraft.
5. Laser Range Finders: Stirling cryocoolers are used in laser range finders to cool the detectors, allowing accurate distance measurements for targeting and artillery applications.
6. Satellite Payloads: Stirling cryocoolers are utilized in space-based satellite payloads for various military purposes, such as reconnaissance, communication, and intelligence gathering. They can cool IR detectors and sensors for space-based surveillance.
7. Remote Sensing: In military applications, remote sensing systems use cooled IR sensors to detect and identify targets from a distance. Stirling cryocoolers play a critical role in cooling these sensors to enhance their performance.
8. Directed Energy Weapons: Stirling cryocoolers are used in high-energy laser weapons to cool the laser medium and optics, ensuring consistent and reliable operation in harsh military environments.
9. Unmanned Aerial Vehicles (UAVs): Miniature Stirling cryocoolers are integrated into UAVs to cool IR sensors, enabling these unmanned platforms to perform reconnaissance and surveillance missions effectively.
10. Secure Communications: Cryogenic cooling can be employed in advanced secure communication systems, such as quantum key distribution (QKD) systems, to improve the security and reliability of military communications.

The use of miniature Stirling cryocoolers in these military applications helps enhance the performance and effectiveness of various defence systems. The cooling technology enables military personnel and equipment to operate with increased precision, accuracy, and

sensitivity, contributing to overall mission success and ensuring the safety and security of military operations.



Fig 3.9: Cryocooler in piston
Ref: https://aerospace.honeywell.com

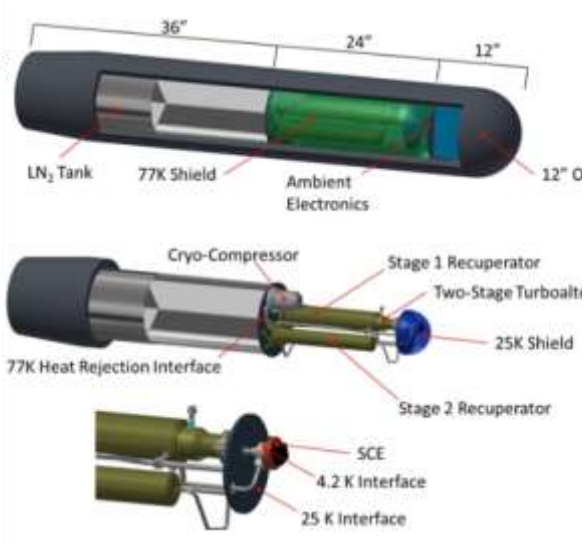
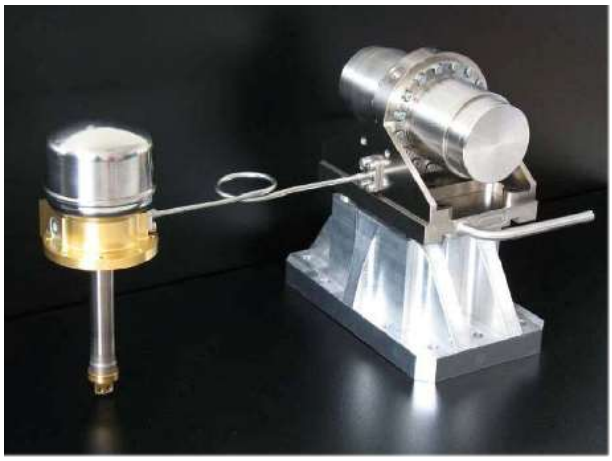


Fig 3.10: 4K **C**ryocooler for tactical applications in military
Ref: www.creare.com

3.10 REFRIGERATION FOR PORTABLE AND REMOTE APPLICATIONS

Due to their compact size and portability, miniature Stirling cryocoolers can be used in remote locations or portable systems, such as field deployable laboratories and military equipment.



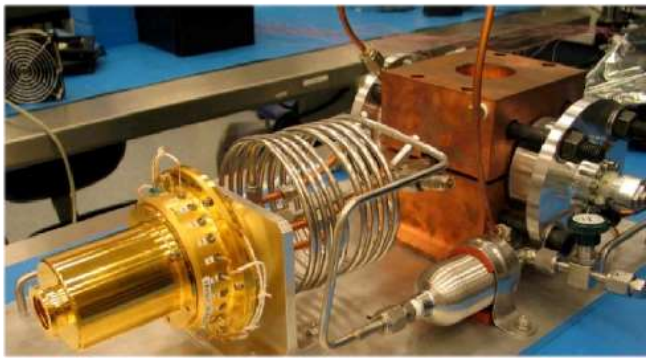


Fig 3.11: Cryocooler for Microsatellite Military Applications
Ref: www.semanticscholar.org

CHAPTER 4: CONCLUSION

As a trainee under Cryogenic department of SSPL, DRDO, I was guided by my mentor to study cryogenics and cryogenic systems. Cryogenic technology plays a crucial role in various scientific, industrial, medical, and military applications. It involves the study and application of extremely low temperatures, typically below -150°C (-238°F), to achieve unique and valuable properties in materials and devices.

Cryogenics has enabled significant advancements and breakthroughs across multiple fields, revolutionizing the way we conduct research, manufacture products, and explore the universe.

Some key points to summarize the importance of cryogenic technology:

- Superconductivity
- Low-Temperature Physics
- Medical Applications
- Industrial Processes
- Space Exploration
- Cooling Solutions

The project on applications of miniature Stirling cryocooler has been carried out successfully. The result obtained is good as expected.

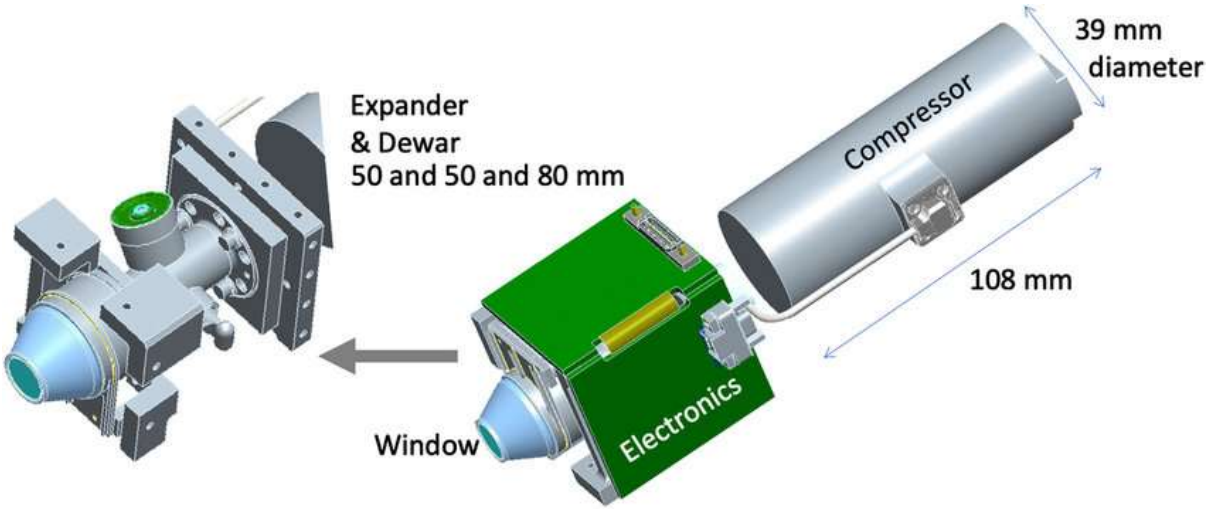


Fig 4.1: Mini Stirling Cryocooler
Ref: www.researchgate.net

CHAPTER 5: FUTURE SCOPE

This project was inspired by the wish to study the different applications of miniature Stirling cryocoolers. The scope of miniature Stirling cryocoolers is vast and holds significant potential in various scientific, industrial, medical, and defence applications. These compact and efficient cooling devices offer several advantages, making them attractive for a wide range of uses. Some of the key areas where miniature Stirling cryocoolers have a promising scope include:

- Portable and Field Applications
- Infrared and Sensor Technologies
- Electronics Cooling
- Space and Satellite Applications
- Quantum Computing
- Cryopreservation and Biomedical Research
- High-Field Science
- Cooling for Military Systems
- Industrial and Manufacturing Processes
- Environmental Monitoring

As technology advances and the demand for compact, reliable, and efficient cooling solutions increases, the scope of miniature Stirling cryocoolers is likely to expand further. Ongoing research and development efforts will lead to improved performance, cost-effectiveness, and broader applications, making miniature Stirling cryocoolers an increasingly valuable tool in a wide range of industries and scientific fields.

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